Project 1: Integration Project
The goal is to find the time it takes an object starting at Earth's orbital radius to fall to the Sun. I used conservation of energy to get $v(R)$ and Simpson's rule to evaluate the integral.

1. Deriving the Integral

$$E_0 = \frac{1}{2}mv_0^2 - \frac{GMm}{R_E}$$
$$\frac{1}{2}mv^2 - \frac{GMm}{R} = E_0.$$

$$\frac{1}{2}v^2 - \frac{GM}{R} = \frac{1}{2}v_0^2 - \frac{GM}{R_E},$$
$$v^2 = v_0^2 + 2GM\left(\frac{1}{R} - \frac{1}{R_E}\right).$$

Since the object is moving toward the Sun,

$$v(R) = -\sqrt{v_0^2 + 2GM\left(\frac{1}{R} - \frac{1}{R_E}\right)}.$$

With $dR/dt = v(R)$,

$$t = \int_{R_E}^{r_S} \frac{1}{v(R)}\,dR = \boxed{\int_{r_S}^{R_E} \frac{dR}{\sqrt{v_0^2 + 2GM\left(\frac{1}{R} - \frac{1}{R_E}\right)}}}.$$

2. Numerical Integration Method

I used the Simpson rule from class:

```
def simpson(f,a,b,N):
    sum = 0.0
    start = float(a)
    step = float(b-a)/N

    for i in range(N):
        sum += (f(start + step*i)
              + 4*f(start + step*(i + 0.5))
              + f(start + step*(i + 1)))*step/6.0

    return sum
```

For $v_0 = 0$, the velocity is zero at R_E, so the original integrand would divide by zero there. I used

$$R = R_E - x^2, \qquad dR = -2x\,dx$$

to avoid that problem. The integral becomes

$$t = \int_0^{\sqrt{R_E - r_S}} \frac{2x\,dx}{\sqrt{v_0^2 + 2GM\left(\frac{1}{R_E - x^2} - \frac{1}{R_E}\right)}}.$$

At $v_0 = 0$ and $x = 0$, I use the limiting value

$$\frac{2R_E}{\sqrt{2GM}}.$$

3. Program

$$R_E = 1.5 \times 10^{11}\ \text{m}, \quad r_S = 7 \times 10^{8}\ \text{m},$$
$$M_S = 2 \times 10^{30}\ \text{kg}, \quad G = 6.67 \times 10^{-11}\ \text{N}\,\text{m}^2/\text{kg}^2.$$

The program uses the Simpson routine above for $v_0 = 0$, 10, and 1000 m/s.

```
import math

RE = 1.5e11
rS = 7.0e8
MS = 2.0e30
G = 6.67e-11

def simpson(f,a,b,N):
    sum = 0.0
    start = float(a)
    step = float(b-a)/N

    for i in range(N):
        sum += (f(start + step*i)
              + 4*f(start + step*(i + 0.5))
              + f(start + step*(i + 1)))*step/6.0

    return sum


def integrand(x,v0):

    if x == 0.0 and v0 == 0.0:
        return 2.0*RE/math.sqrt(2.0*G*MS)

    R = RE - x*x

    return 2.0*x/math.sqrt(
        v0*v0 + 2.0*G*MS*(1.0/R - 1.0/RE)
    )


N = 10000
a = 0.0
b = math.sqrt(RE-rS)

for v0 in [0.0, 10.0, 1000.0]:
    t = simpson(lambda x: integrand(x,v0),a,b,N)
    print(v0,t,t/86400.0)
```

4. Results

v_0 (m/s)	Time (s)	Time (days)
0	5.58605×10^6	64.6534
10	5.58437×10^6	64.6339
1000	5.42198×10^6	62.7543

The 0 and 10 m/s cases are very close because the initial 10 m/s is small compared with the speeds reached later.

5. Questions

(1) What trouble do you run into in trying to evaluate this integral for $v_0 = 0$?

At the starting point,

$$v(R_E) = -\sqrt{2GM\left(\frac{1}{R_E} - \frac{1}{R_E}\right)} = 0.$$

So the original integrand has a division by zero at R_E. I used the $R = R_E - x^2$ substitution above so Simpson's rule can still be used.

(2) Our formulation assumed that v_0 was directed toward the Sun. How would it change if it was directed away?

If v_0 points away from the Sun, R first increases. The object reaches a maximum radius where $v = 0$:

$$0 = v_0^2 + 2GM\left(\frac{1}{R_{\max}} - \frac{1}{R_E}\right),$$

$$\boxed{R_{\max} = \frac{1}{\frac{1}{R_E} - \frac{v_0^2}{2GM}}}.$$

The object then turns around and falls toward the Sun. The total time is the outward trip to $R_{\max}$ plus the return trip to r_S. If the initial speed is at least the escape speed, there is no finite $R_{\max}$ and the object does not return in this model.

(3) Using Newton's Laws, set up a differential equation for $r(t)$ directly.

$$F = -\frac{GMm}{R^2} = m\frac{d^2R}{dt^2}$$

$$\boxed{\frac{d^2R}{dt^2} = -\frac{GM}{R^2}}$$

$$\boxed{R(0) = R_E, \qquad R'(0) = -v_0.}$$